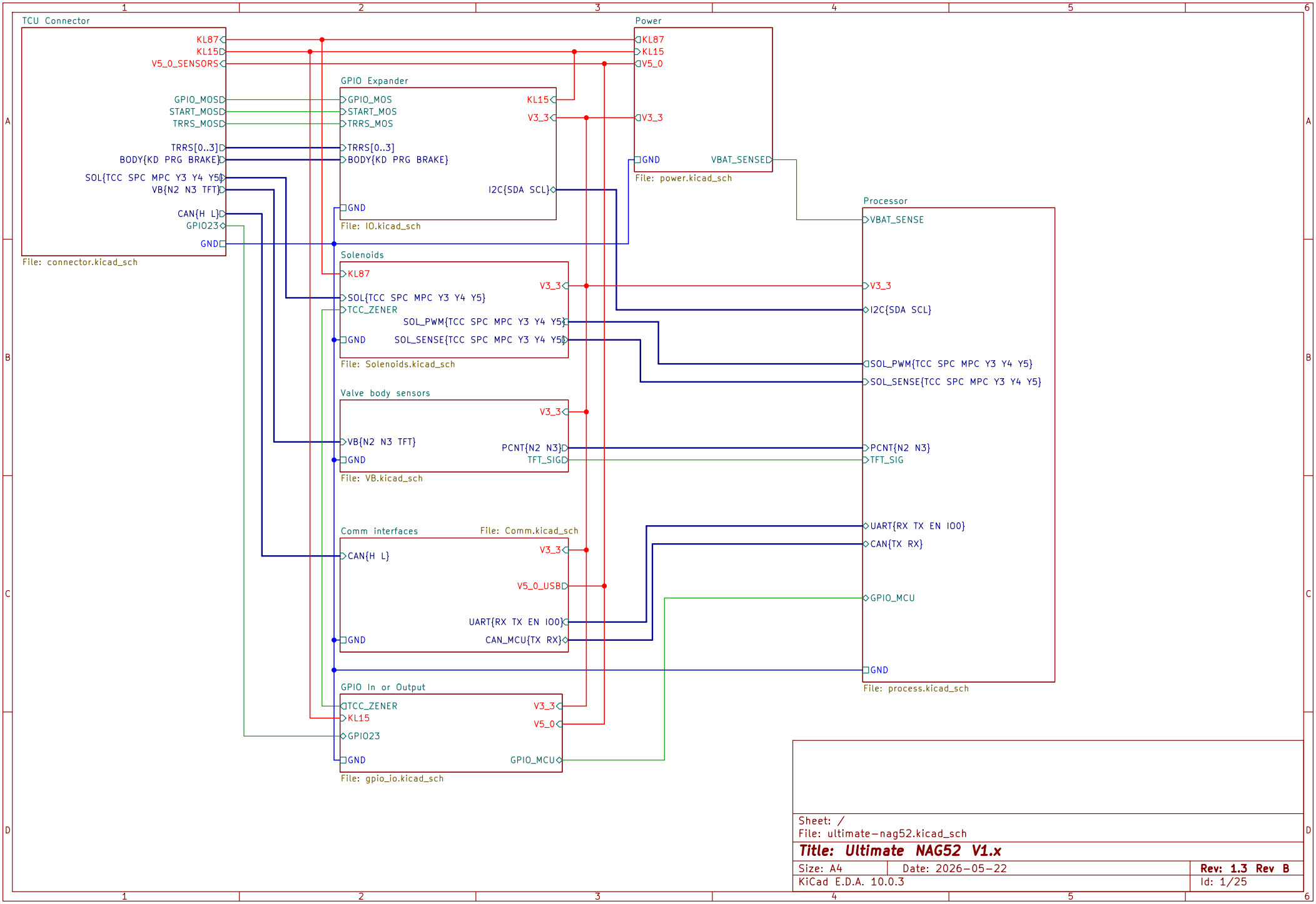
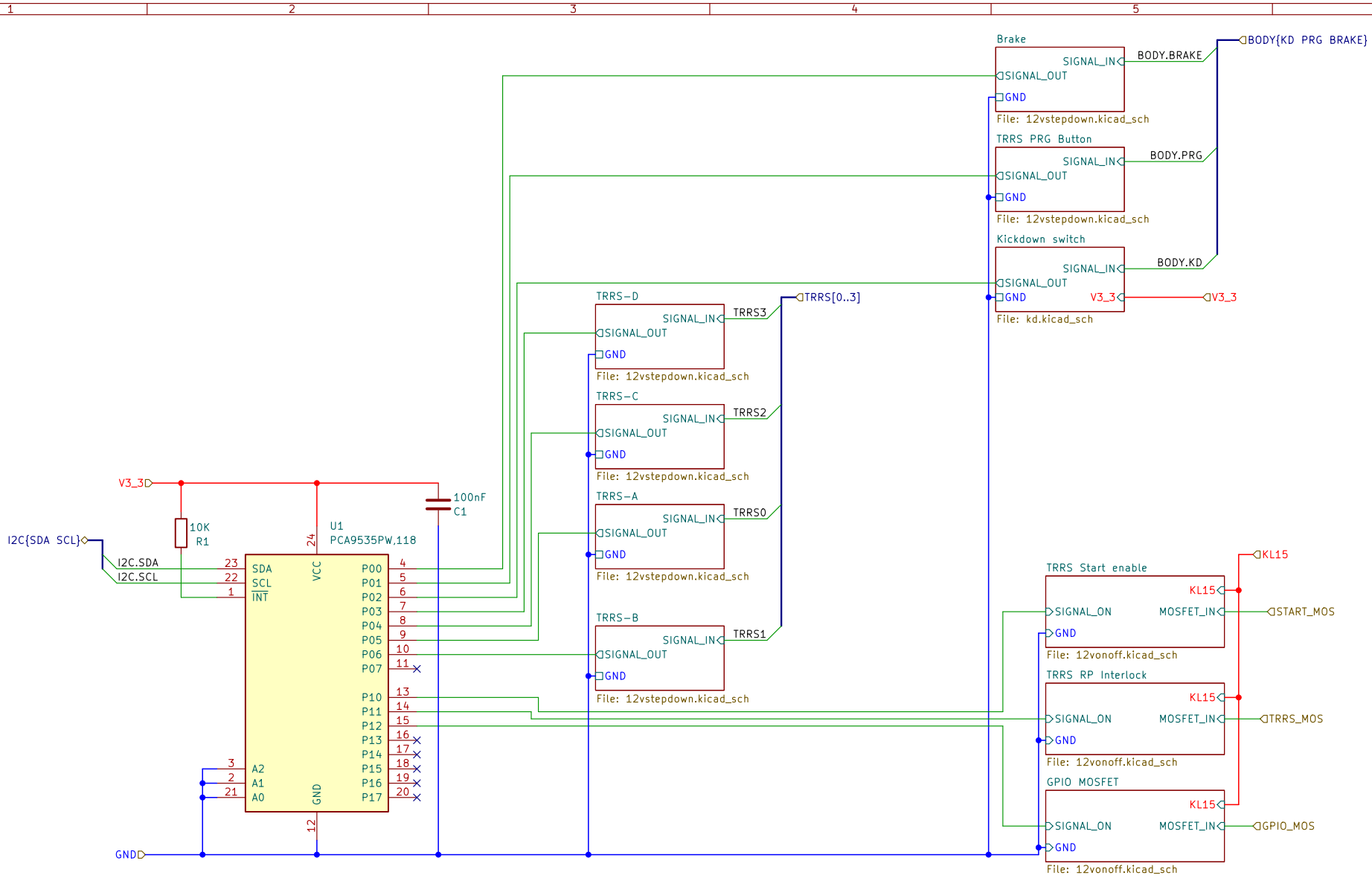
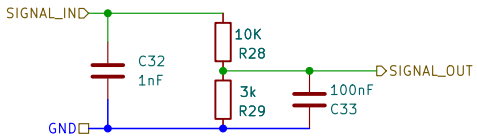

TCU Connector
KL87
KL15
V5_0_SENSORS
GPIO_MOS
START_MOS
TRRS_MOS
TRRS[0..3]
BODY{KD PRG BRAKE}
SOL{TCC SPC MPC Y3 Y4 Y5}
VB{N2 N3 TFT}
CAN{H L}
GPIO23
GND
File: connector.kicad_sch
GPIO Expander
GPIO_MOS
START_MOS
TRRS_MOS
TRRS[0..3]
BODY{KD PRG BRAKE}
GND
KL15
V3_3
I2C{SDA SCL}
File: IO.kicad_sch
Power
KL87
KL15
V5_0
V3_3
GND
VBAT_SENSED
File: power.kicad_sch
Processor
VBAT_SENSE
V3_3
I2C{SDA SCL}
SOL_PWM{TCC SPC MPC Y3 Y4 Y5}
SOL_SENSE{TCC SPC MPC Y3 Y4 Y5}
PCNT{N2 N3}
TFT_SIG
UART{RX TX EN IO0}
CAN{TX RX}
GPIO_MCU
GND
File: process.kicad_sch
Solenoids
KL87
SOL{TCC SPC MPC Y3 Y4 Y5}
TCC_ZENER
GND
V3_3
SOL_PWM{TCC SPC MPC Y3 Y4 Y5}
SOL_SENSE{TCC SPC MPC Y3 Y4 Y5}
File: Solenoids.kicad_sch
Valve body sensors
VB{N2 N3 TFT}
GND
V3_3
PCNT{N2 N3}
TFT_SIG
File: VB.kicad_sch
Comm interfaces
File: Comm.kicad_sch
CAN{H L}
GND
V3_3
V5_0_USB
UART{RX TX EN IO0}
CAN_MCU{TX RX}
GPIO In or Output
TCC_ZENER
KL15
GPIO23
GND
V3_3
V5_0
GPIO_MCU
File: gpio_io.kicad_sch
Sheet: /
File: ultimate-nag52.kicad_sch
Title: Ultimate NAG52 V1.x
Size: A4
Date: 2026-05-22
Rev: 1.3 Rev B
KiCad E.D.A. 10.0.3
Id: 1/25



Sheet: /GPIO Expander/
File: IO.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

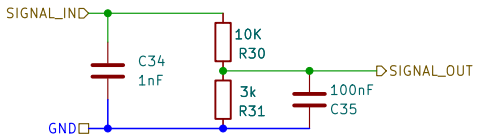
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 2/25



Sheet: /GPIO Expander/TRRS-A/
File: 12vstepdown.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

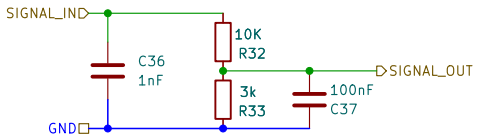
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 10/25



Sheet: /GPIO Expander/TRRS-B/
File: 12vstepdown.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

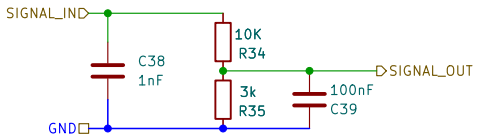
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 11/25



Sheet: /GPIO Expander/TRRS-C/
File: 12vstepdown.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

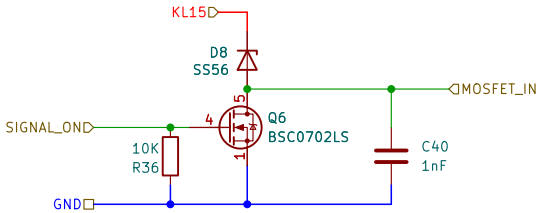
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 12/25



Sheet: /GPIO Expander/TRRS-D/
File: 12vstepdown.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

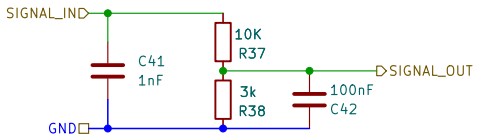
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 13/25



Sheet: /GPIO Expander/TRRS Start enable/
File: 12vonoff.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

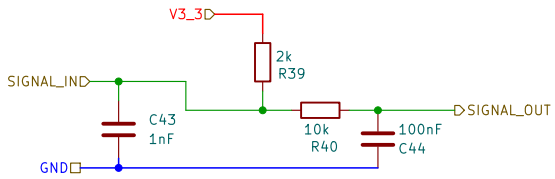
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 14/25



Sheet: /GPIO Expander/Brake/
File: 12vstepdown.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

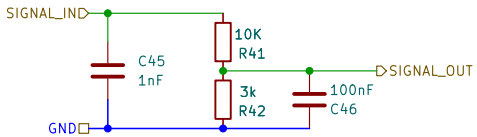
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 15/25



Sheet: /GPIO Expander/Kickdown switch/
File: kd.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

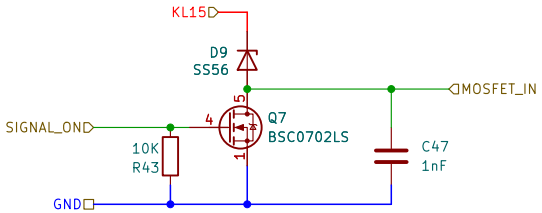
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 16/25



Sheet: /GPIO Expander/TRRS PRG Button/
File: 12vstepdown.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

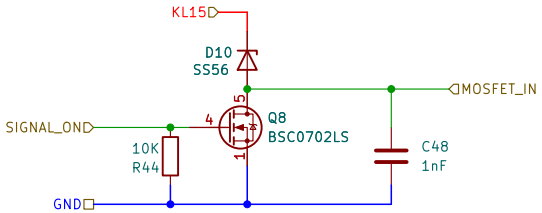
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KiCad E.D.A. 10.0.3		Id: 17/25



Sheet: /GPIO Expander/GPIO MOSFET/
File: 12vonoff.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

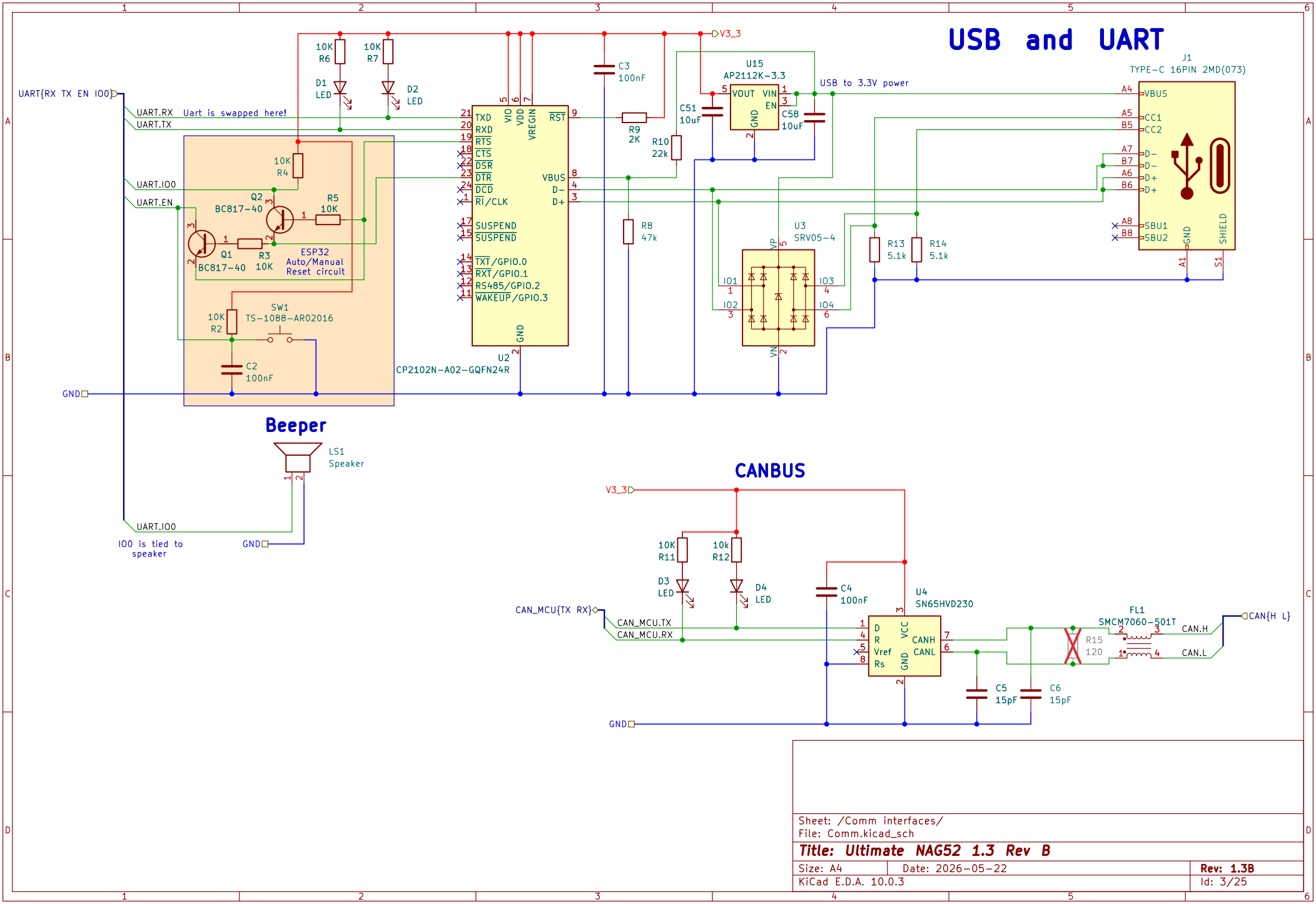
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 18/25



Sheet: /GPIO Expander/TRRS RP Interlock/
File: 12vonoff.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 19/25



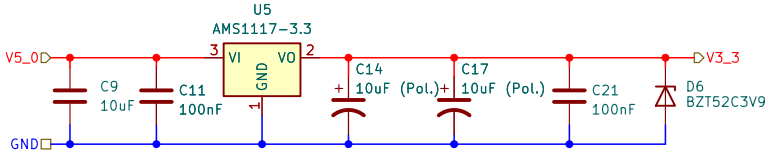
Sheet: /Comm interfaces/
File: Comm.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 3/25

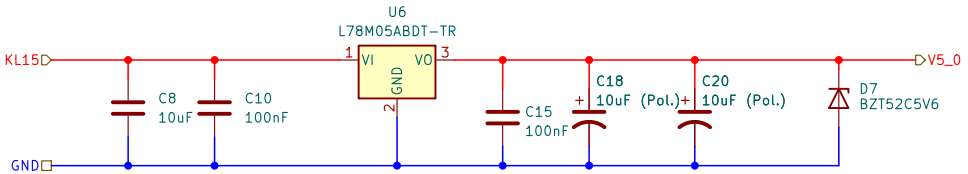
5.0V to 3.3V

Handles up to 1A

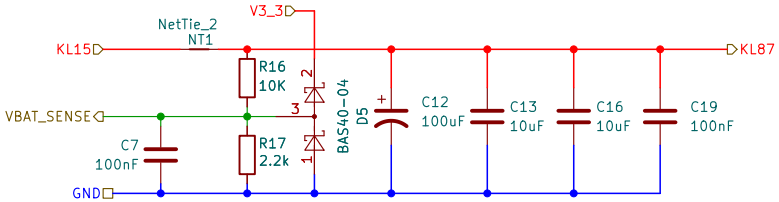


PCB 5.0V Supply

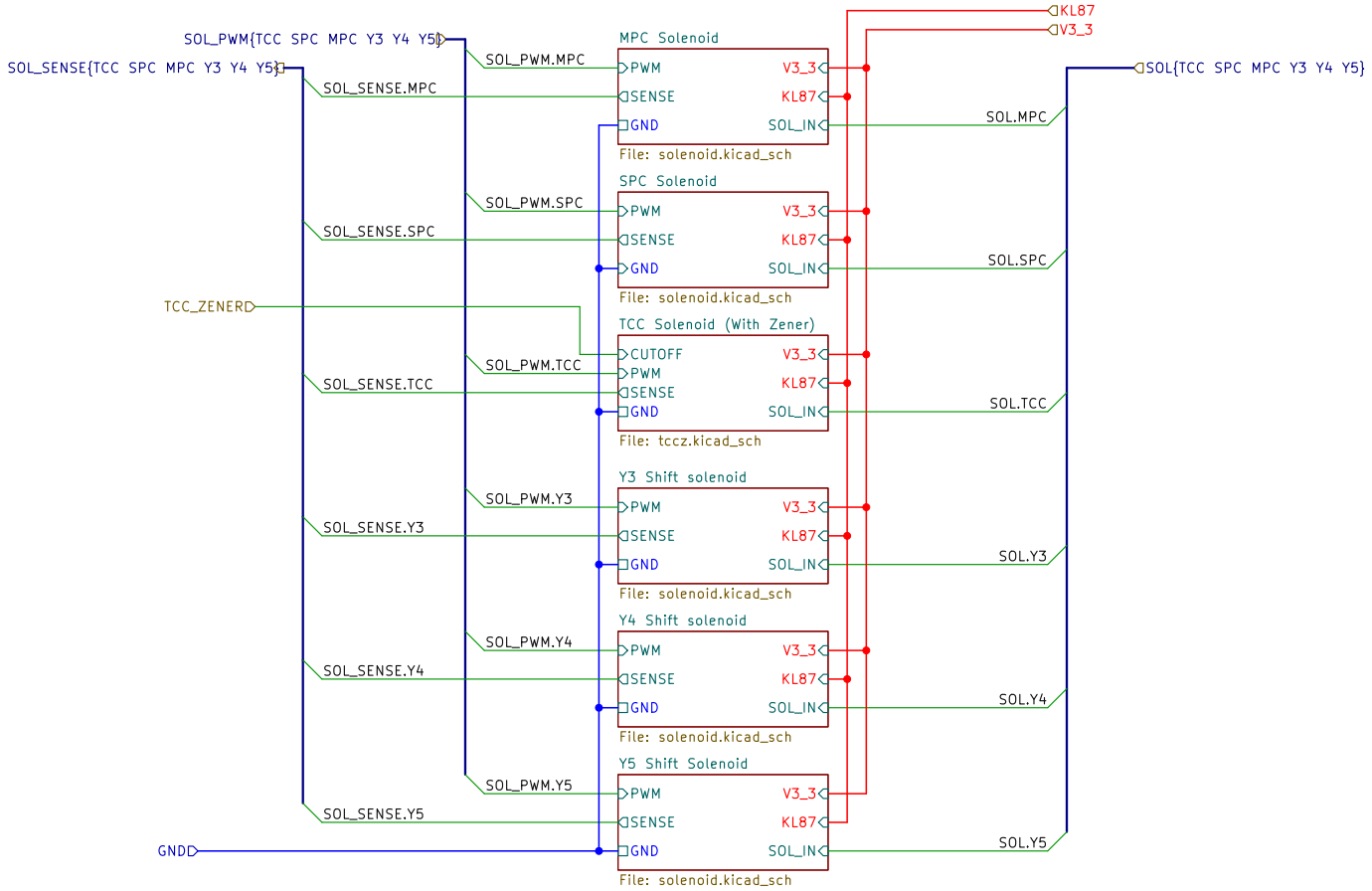
Handles up to 0.5A



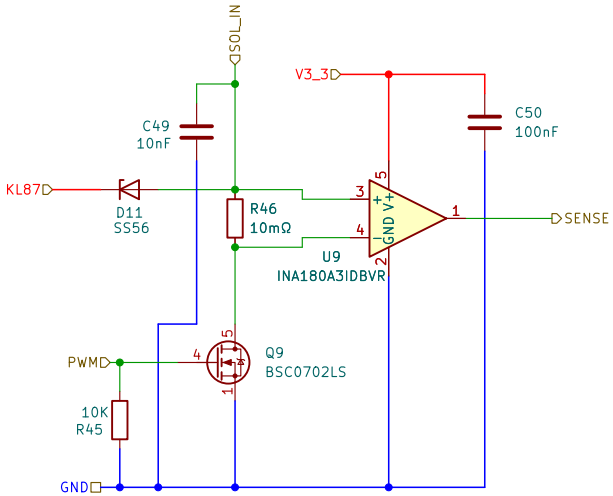
Solenoid power and monitoring



Sheet: /Power/		
File: power.kicad_sch		
Title: Ultimate NAG52 1.3 Rev B		
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 4/25



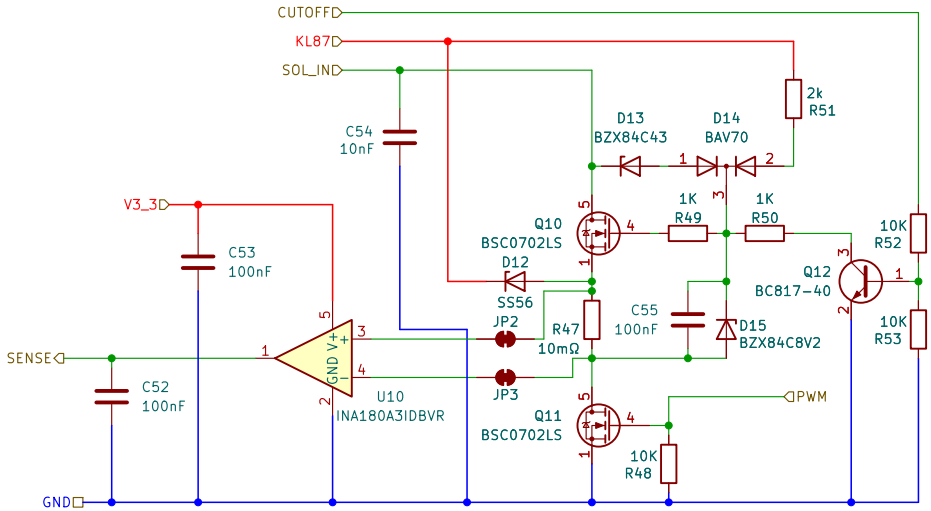
Sheet: /Solenoids/		
File: Solenoids.kicad_sch		
Title: Ultimate NAG52 1.3 Rev B		
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 5/25



Sheet: /Solenoids/MPC Solenoid/
File: solenoid.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

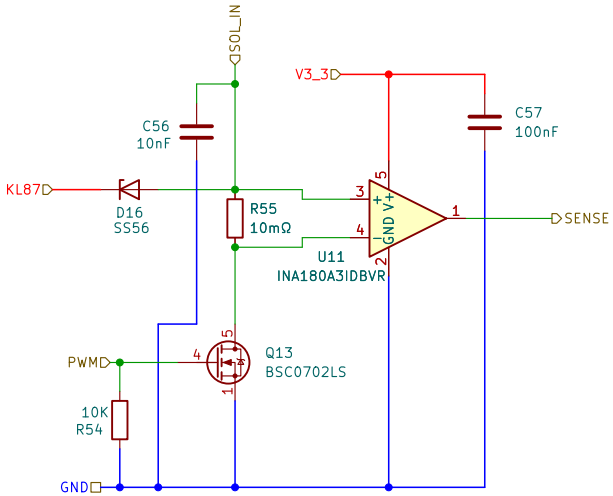
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 20/25



Sheet: /Solenoids/TCC Solenoid (With Zener)/
File: tccz.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

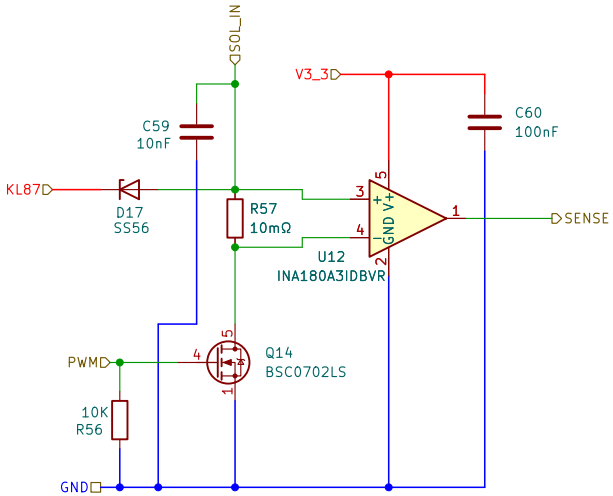
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 21/25



Sheet: /Solenoids/Y3 Shift solenoid/
File: solenoid.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

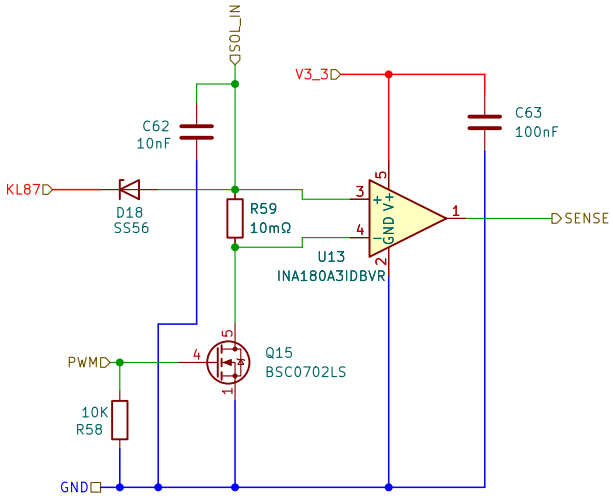
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 22/25



Sheet: /Solenoids/Y4 Shift solenoid/
File: solenoid.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

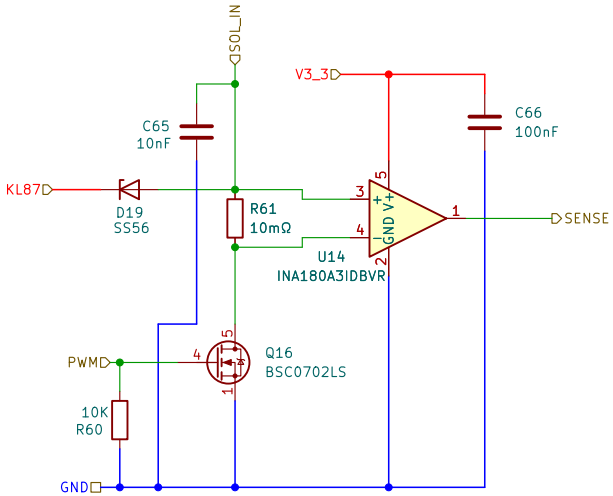
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 23/25



Sheet: /Solenoids/Y5 Shift Solenoid/
File: solenoid.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

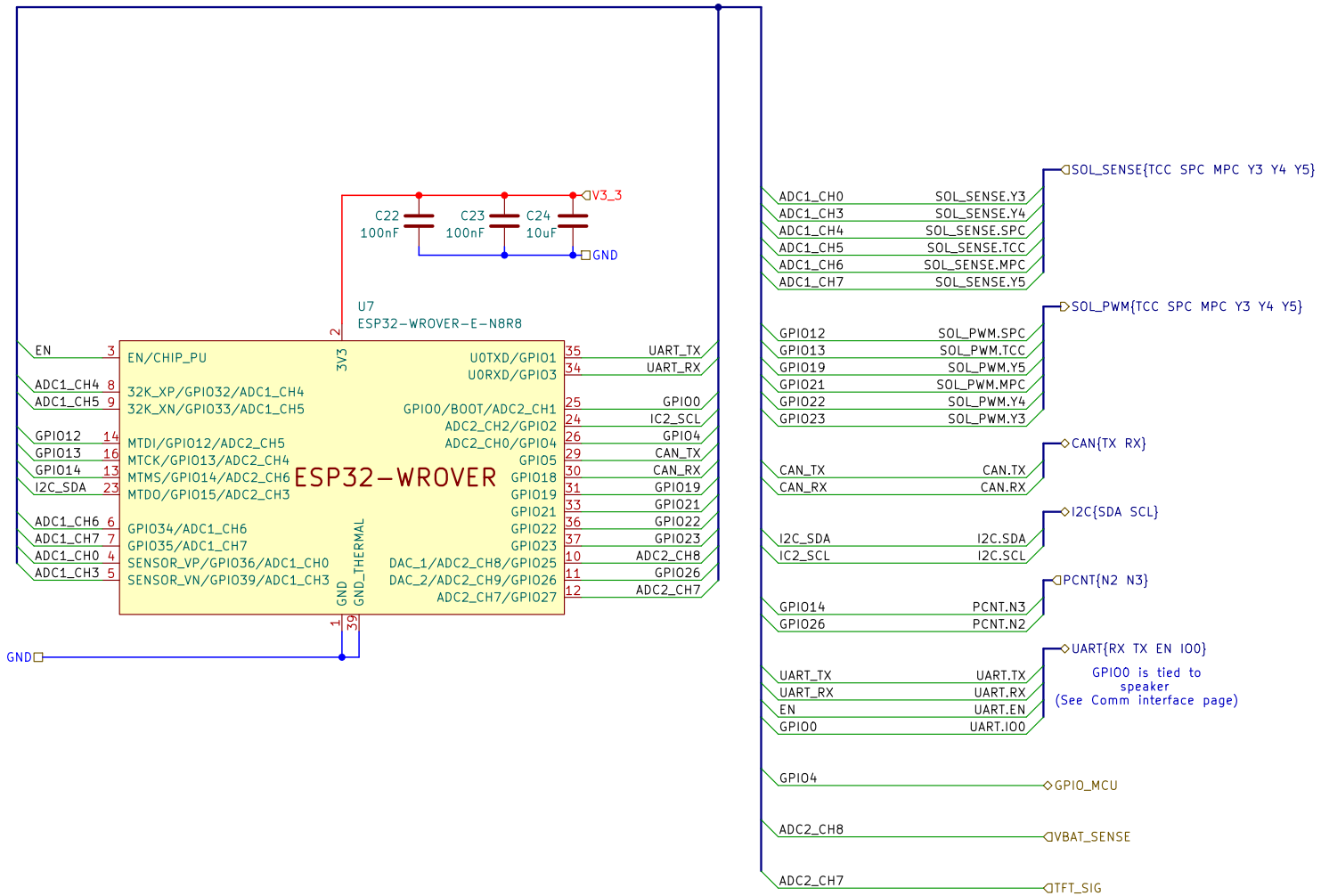
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 24/25



Sheet: /Solenoids/SPC Solenoid/
File: solenoid.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

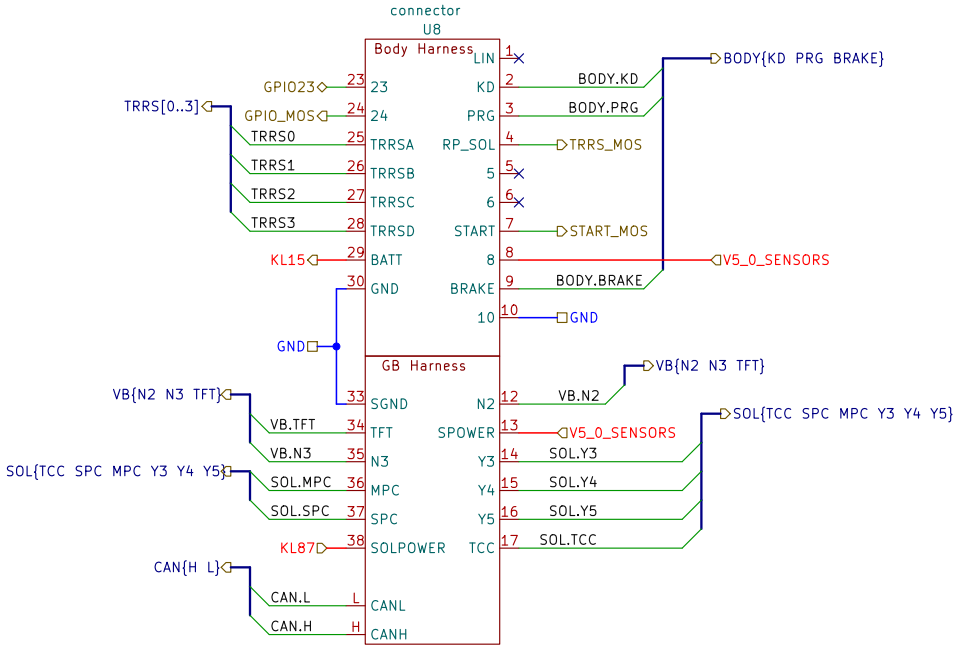
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 25/25



Sheet: /Processor/
File: process.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

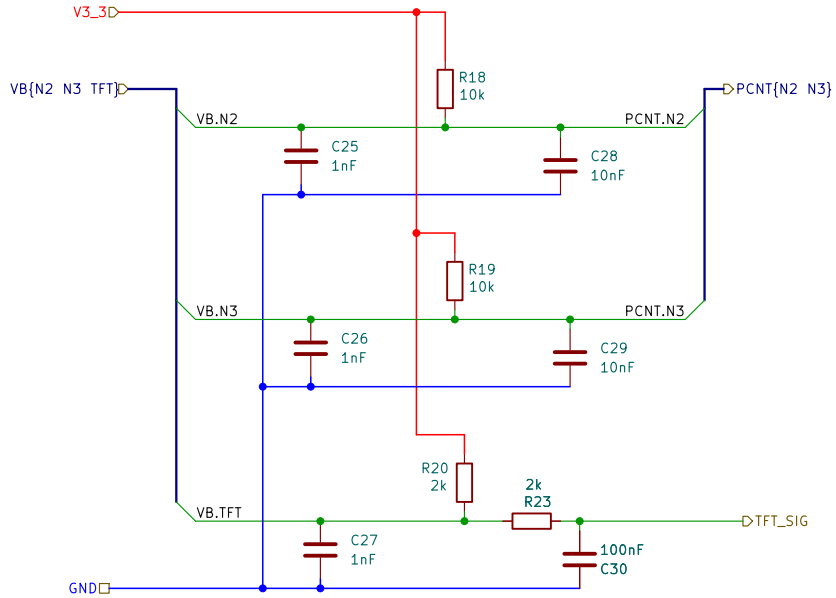
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KiCad E.D.A. 10.0.3		Id: 6/25



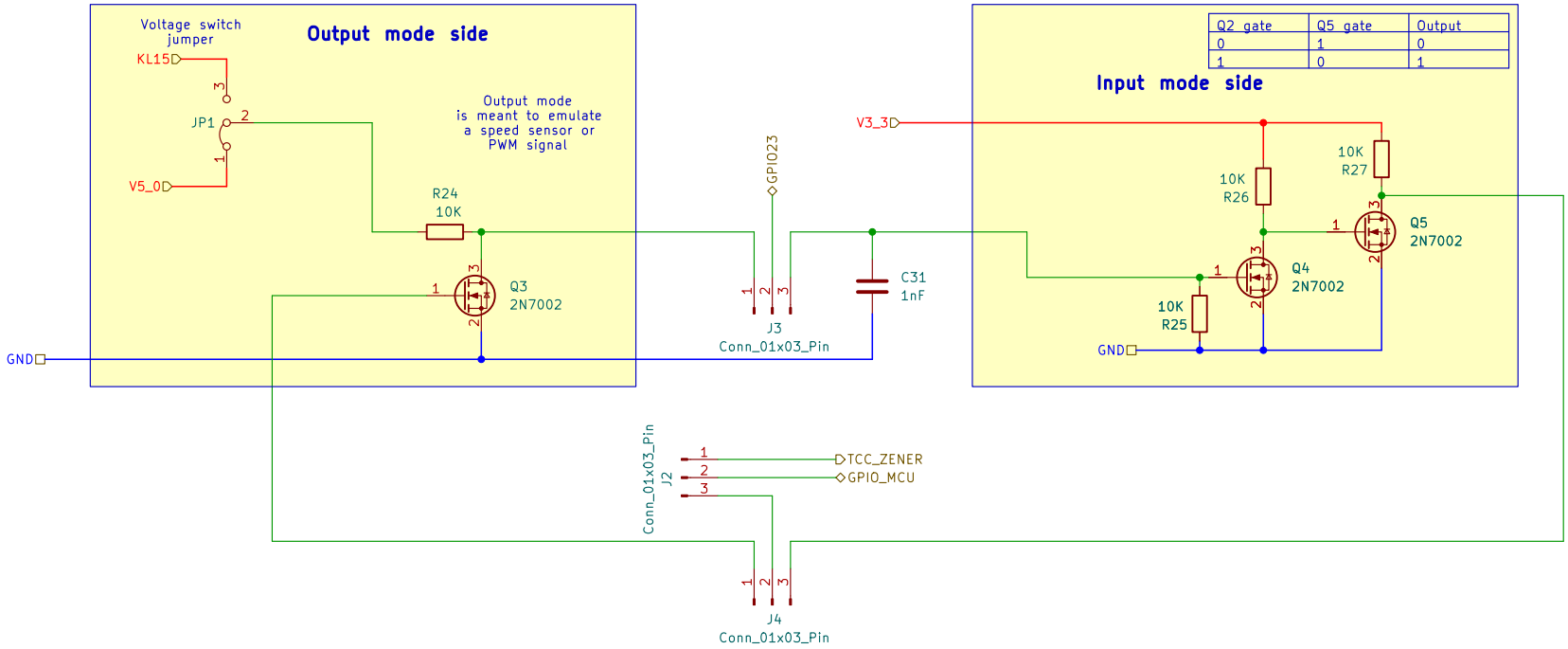
Sheet: /TCU Connector/
File: connector.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 7/25



Sheet: /Valve body sensors/		
File: VB.kicad_sch		
Title: Ultimate NAG52 1.3 Rev B		
Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 8/25



Sheet: /GPIO In or Output/
File: gpio_io.kicad_sch

Title: Ultimate NAG52 1.3 Rev B

Size: A4	Date: 2026-05-22	**Rev: 1.3B**
KiCad E.D.A. 10.0.3		Id: 9/25